

02SXJJM – Fundamentals of Machine Design –
A.Y. 2023/2024

Mahdy Mokdad, Mohammad Mahdi, Mohammad Nour edeen

January 2024



**Politecnico
di Torino**

Presented to Professor **Nicola Bosso**
and co-staffs **Matteo Magelli** and **Francesco Mocera**

Summary

1	Introduction	2
2	Static Verification	2
2.1	Forces on Gears	2
2.2	Forces on Shaft Ends	3
2.3	Internal Reactions	5
2.4	Internal Stresses	6
2.5	Equivalent Stresses and safety factor	7
3	Fatigue Verification	8
3.1	Evaluation of alternate and mean stress on Shaft A2	8
3.2	Fatigue Coefficients	8
3.3	Fatigue limit correction factors	9
3.4	Fatigue safety factor	11
4	Gear Tooth Verification	13
4.1	Maximum tooth gear bending stress equation for fatigue	13
4.2	Bending Safety Factor	18
4.3	Maximum gear contact (pitting resistance) stress equation in fatigue	18
4.4	Wear safety factor (Hertzian contact)	20
5	Bearings Life Investigation	20
5.1	Fatigue Life Bearing	20
5.2	Static Failure	23
6	Table For Key Calculation Results	24

List of Tables

1	Normal Stresses	6
2	Bending Stresses	7
3	Bending Stresses	7
4	Torsion Stresses	7
5	Resulting Normal Stress	8
6	Safety factors for different cross sections	8
7	Alternate and Mean Stress Components at important cross sections	8
8	Stress Concentration Coefficients	9
9	Safety factors against fatigue accompanied with the stresses of interest	11
10	C and C_0	23
11	Summary for all data concerning bearing analysis	23

1 Introduction

In this project we are going to analyze a gearbox in what regards the static and fatigue life of the helical gears as well as for the bearings contained in it. We start up by doing preliminary calculations regarding powers and torques. Following the dissection of each cross section.

2 Static Verification

2.1 Forces on Gears

We begin by calculating the forces on each Gear and then we move on to calculate the reactions:

$$\begin{aligned} T_m &= \frac{P_m}{\omega} \quad [\text{N m}] \\ &= \frac{36000}{50\pi} = 229.18 \text{ N m} \end{aligned} \tag{1}$$

The forces on gear 1 and gear 2:

$$F_{t1} = F_{t2} = \frac{T_m}{r_1} = 9223.9 \text{ N} \tag{2}$$

$$F_{r1} = F_{r2} = F_{t1} \cdot \tan(\phi_t) = 3475.658 \text{ N} \tag{3}$$

$$F_{a1} = F_{a2} = F_{t1} \cdot \psi = 2471.53 \text{ N} \tag{4}$$

For calculating the forces at gear 4 of the output shaft, we calculate the torque first:

$$\omega_{out} = \frac{\omega_{in}}{i} = 17.453 \text{ rad/sec} \tag{5}$$

$$T_{out} = \frac{P_m}{\omega_{out}} = 2062.64 \text{ N m} \tag{6}$$

Now the forces are:

$$F_{t4} = F_{t3} = \frac{T_{out}}{r_4} = 15\,812.75 \text{ N} \tag{7}$$

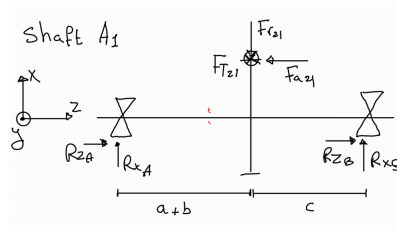
In the same manner equation(3) and equation(4), we calculate radial and axial force:

$$F_{r4} = F_{r3} = 5956.227 \text{ N}$$

$$F_{a4} = F_{a3} = 4237 \text{ N}$$

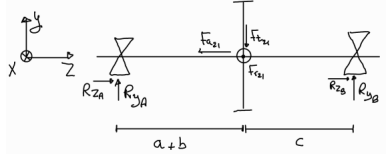
2.2 Forces on Shaft Ends

Now consider Shaft A1 and balancing out forces we find out the horizontal and vertical forces acting on opposite ends(where bearings are located):



$$R_{xB} = 2195 \text{ N}$$

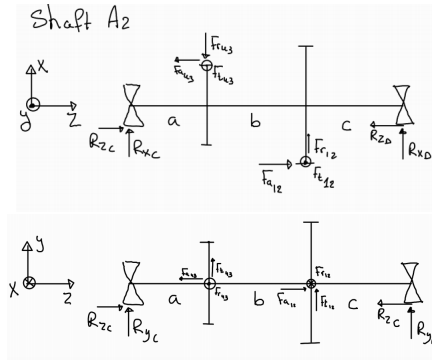
$$R_{xA} = 1280.658 \text{ N}$$



$$R_{yB} = 6594.81 \text{ N}$$

$$R_{yA} = 2629.1 \text{ N}$$

Similarly, for shafts A2 and A3:

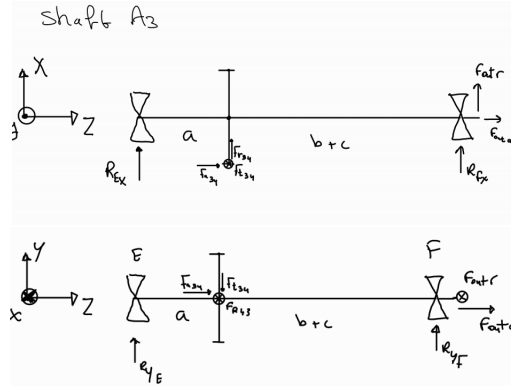


$$R_{xD} = -1999.865 \text{ N}$$

$$R_{xC} = 4480.434 \text{ N}$$

$$R_{yD} = -12\,590.09 \text{ N}$$

$$R_{yC} = -12\,446.57 \text{ N}$$



One additional step is calculating $F_{out r}$ and $F_{out t}$:

$$F_{out r} = \frac{T_{A3}}{0,25} = 8250.59 \text{ N}$$

$$F_{out t} = 0,25 \cdot F_{out r} = 2062.64 \text{ N}$$

$$R_{xF} = -16\,134.164 \text{ N}$$

$$R_{xE} = 2107.347 \text{ N}$$

$$R_{yF} = 5984.226 \text{ N}$$

$$R_{yE} = 9828.52 \text{ N}$$

Moving to the axial Reaction forces:

$$R_{zA} = 0 \text{ N}$$

$$R_{zB} = 2472.53 \text{ N}$$

$$R_{zC} = 0 \text{ N}$$

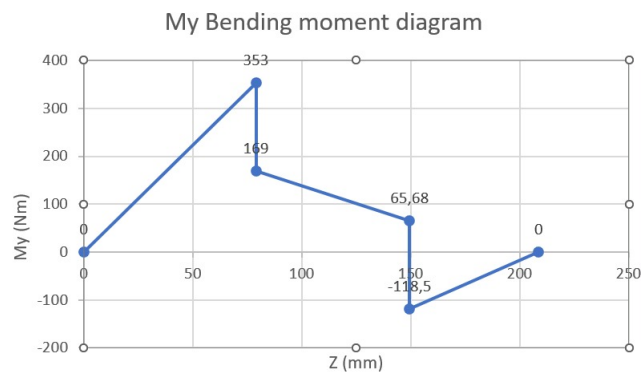
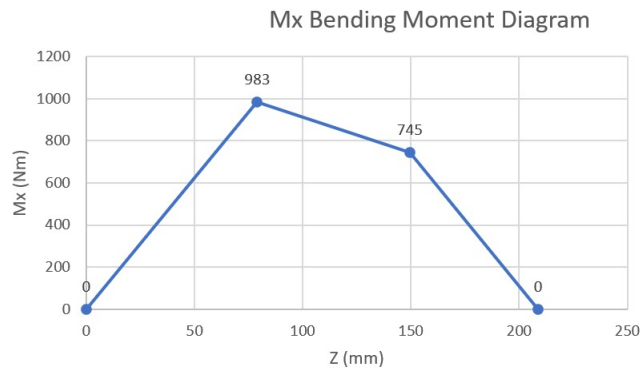
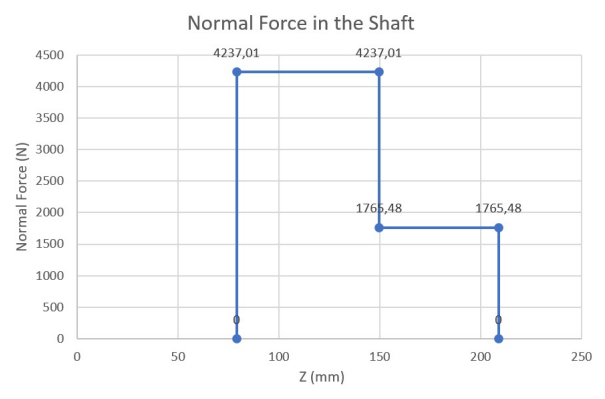
$$R_{zD} = 1765.48 \text{ N}$$

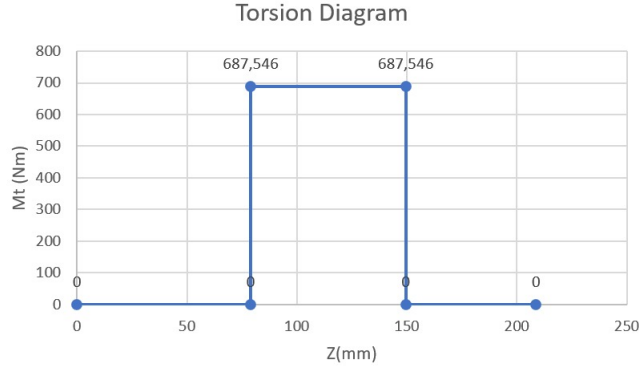
$$R_{zE} = 0 \text{ N}$$

$$R_{zF} = 6299 \text{ N}$$

2.3 Internal Reactions

Considering that shaft A2 is the most critical component in our system we calculate internal loads and verify the safety factors. Below are the diagrams for the internal reactions:





These diagrams are obtained by balancing the forces and moments at each cross-section.

2.4 Internal Stresses

Following the Internal reactions calculated, we can calculate the stresses at the cross section of concern. Starting with σ^N which is calculated according to:

$$\sigma^N = \frac{N}{A} \quad [\text{MPa}] \quad (8)$$

Cross Section	σ^N
V_1	0.9192 MPa
V_2	1.403 MPa
V_3	1.72 MPa
V_4	2.24 MPa

Table 1: Normal Stresses

Moving on to σ^{Mb} :

$$\sigma^{Mb} = \frac{\sqrt{M_x^2 + M_y^2}}{W_b} \quad [\text{MPa}] \quad (9)$$

Cross Section	σ^{Mb}
V_1	23.67 MPa
V_2	38.39 MPa
V_3	47.74 MPa
V_4	65.39 MPa

Table 2: Bending Stresses

Cross Section	σ^{Mb}
V_1	23.67 MPa
V_2	38.39 MPa
V_3	47.74 MPa
V_4	65.39 MPa

Table 3: Bending Stresses

Finally τ^{Mt} :

$$\tau^{Mt} = \frac{M_t}{W_t} \quad [\text{MPa}] \quad (10)$$

Cross Section	τ^{Mt}
V_1	7.78 MPa
V_2	14.692 MPa
V_3	19.93 MPa
V_4	29.76 MPa

Table 4: Torsion Stresses

2.5 Equivalent Stresses and safety factor

Second step includes the calculation is the equivalent stress at each cross section following the Shigley's Formula:

$$\sigma^{tot} = \sigma^{Mb} + \sigma^N \quad (11)$$

Cross Section	σ^{tot}
V ₁	23.67 MPa
V ₂	38.39 MPa
V ₃	47.74 MPa
V ₄	65.39 MPa

Table 5: Resulting Normal Stress

$$\sigma_{id} = \sqrt{(\sigma^{Mb} + \sigma^N)^2 + 3(\tau^{Mt})^2} \quad [\text{MPa}] \quad (12)$$

$$SF = \frac{\sigma_y}{\sigma_{id}} \quad (13)$$

In conclusion section V4 is the most critical cross-section for static failure,

Cross Section	σ_{id}	SF
V ₁	28,039	33,s16
V ₂	46,931	19,81
V ₃	60,31	15,42
V ₄	75,93	12,24

Table 6: Safety factors for different cross sections

however as the SF is high, it wouldn't be much of a concern.

3 Fatigue Verification

3.1 Evaluation of alternate and mean stress on Shaft A2

For each cross section we evaluate the mean and alternate stress components due to normal, bending and torsional loads. The values tabulated are :

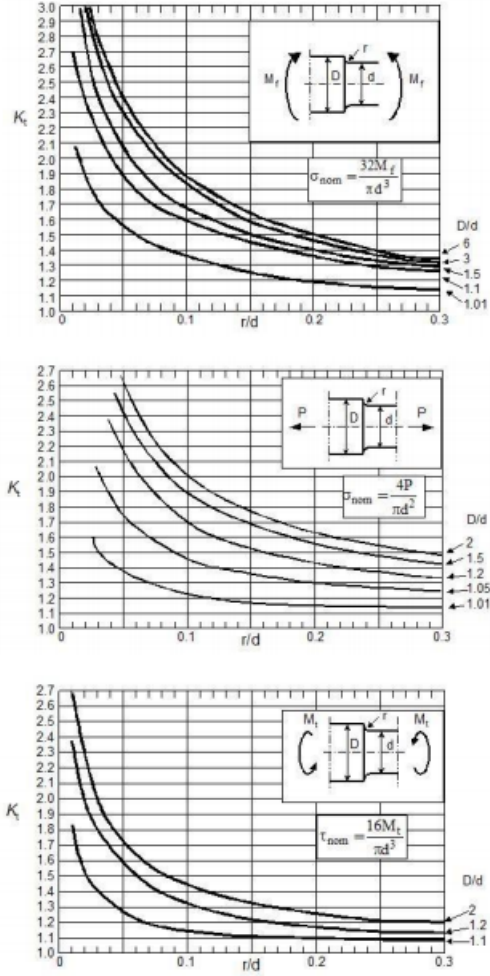
Cross section	σ_m^N	σ_a^N	σ_m^{Mb}	σ_a^{Mb}	τ_m^{Mt}	τ_a^{Mt}
V1	0,9192	0	0	23,67	7,78	0
V2	1,403	0	0	38,39	14,62	0
V3	1,72	0	0	47,74	19,93	0
V4	1,72	0	0	43,80	19,93	0

Table 7: Alternate and Mean Stress Components at important cross sections

3.2 Fatigue Coefficients

Due to the changes in geometry, intensification factors are present on different kinds of stresses. These stress raiser factors sometimes deteriorate the fatigue life of the component. To take these into consideration the following coefficients are evaluated.

Geometrical stress concentration factors K_t



Cross-section	q	K_{tN}	K_{fN}	K_{tB}	K_{fB}	K_{tT}	K_{fT}
V1	-	-	1	-	1,2	-	1,5
V2	0,9	2,5	2,35	2,28	2,152	1,8	1,72
V3	0,88	2,4	2,232	2,5	2,32	1,6	1,528
V4	-	-	1	-	1,6	-	2

Table 8: Stress Concentration Coefficients

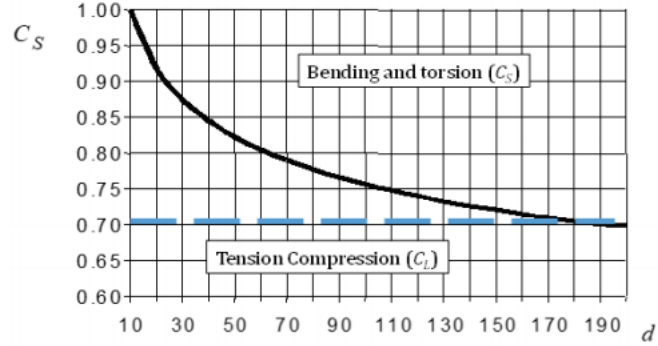
3.3 Fatigue limit correction factors

In order to calculate the safety factor for fatigue we proceed using the Haigh diagram identifying the infinite life fatigue strength (σ_{D-1}^C) and the working

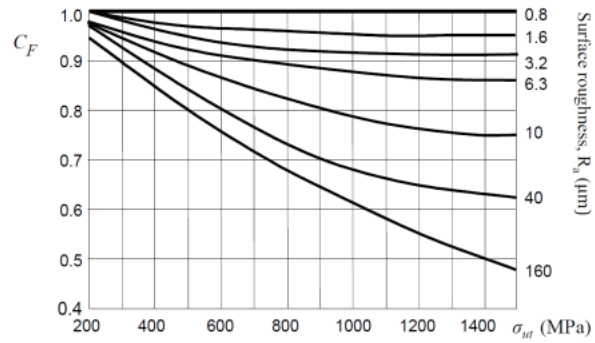
conditions.

σ_{D-1}^C is calculated using the infinite life for a specimen which is σ_{D-1} multiplied by Size and surface finish factors.

Size Effect C_S



Surface finish effect



These factors are extracted from the following graphs knowing the values of the diameter roughness and the ultimate strength of the shaft material in use. The Surface finish effect factor is

$$C_F = 0,955$$

and it is equal for all 4 cross-sections.

And the values of C_s are:

$$C_{s1} = 0,78; C_{s2} = 0,8; C_{s3} = 0,82; C_{s4} = 0,82$$

3.4 Fatigue safety factor

Based on our findings the infinite fatigue life of our component is calculated:

$$\sigma_{D-1}^C = \sigma_{D-1} \cdot C_F \cdot C_s$$

Now the equivalent alternate and mean stress components should be calculated according to the following:

$$\sigma_a^{eq} = \sqrt{(K_f^B \cdot \sigma_a^B + \frac{K_f^N \cdot \sigma_a^B}{0,85})^2 + 3(K_f^T \cdot \tau_a^T)^2} \quad (14)$$

$$\sigma_m^{eq} = \sqrt{(K_f^B \cdot \sigma_m^B + K_f^N \cdot \sigma_m^B)^2 + 3(K_f^T \cdot \tau_m^T)^2} \quad (15)$$

And finally the safety factor is calculated according to:

$$\frac{1}{SF} = \frac{\sigma_a^P}{\sigma_{D-1}^C} + \frac{\sigma_m^P}{\sigma_u} \quad (16)$$

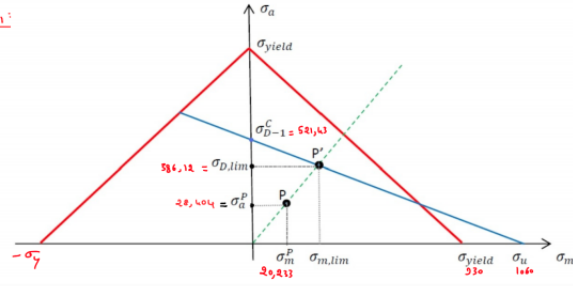
Cross Section	σ_{D-1}^C	σ_a^{eq}	σ_m^{eq}	SF
V1	521,43	28,404	20,33	13,59
V2	534,8	82,615	43,893	5,1
V3	548,17	110,756	52,885	3,969
V4	548,17	70,09	69,13	5,179

Table 9: Safety factors against fatigue accompanied with the stresses of interest

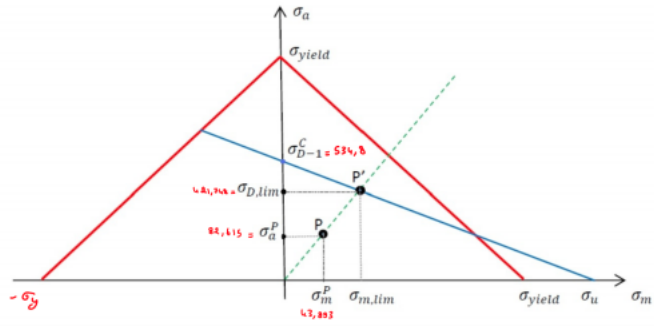
Turned out the cross section V3 is the most prone cross section to fail under fatigue.

Below are the haigh diagrams for all cross sections:

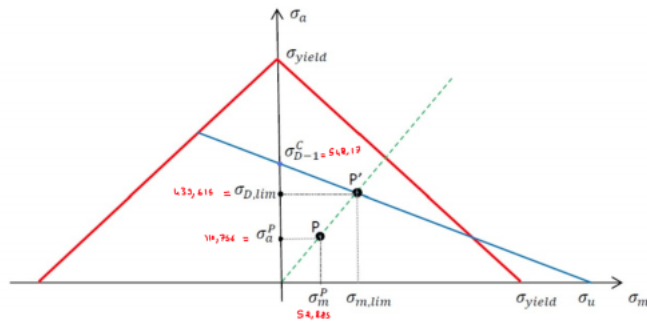
V₁:

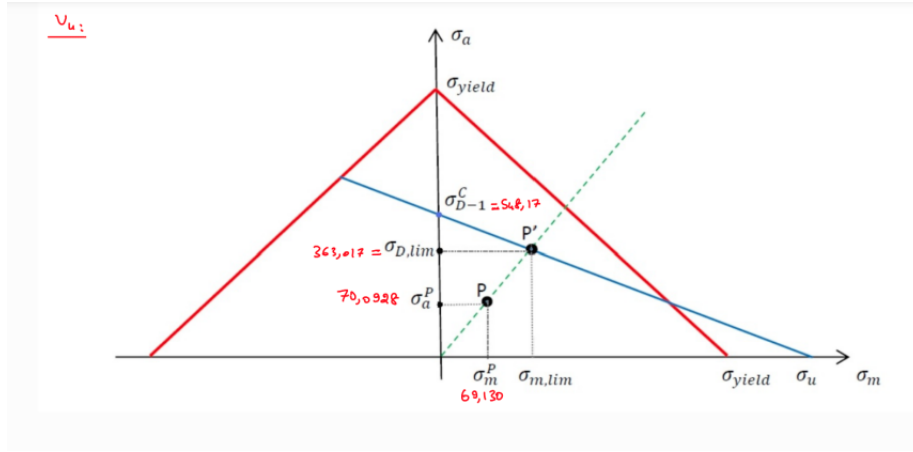


V₂:



V₃:





4 Gear Tooth Verification

In this section we will focus on verifying the teeth of pinion G3 which will be the most critical gear for our analysis.

4.1 Maximum tooth gear bending stress equation for fatigue

Starting from the following equation and after calculating the proper coefficients the maximum tooth gear bending stress

$$\sigma_{max, Fatigue} = F_t K_0 K_B K_V K_H K_S \frac{1}{b \cdot m_t Y_J} \quad (17)$$

- **Face width b :**

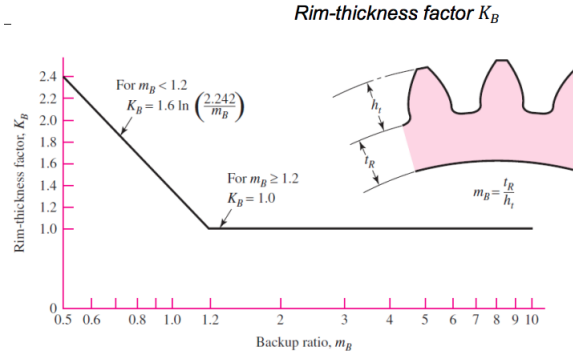
$$b = \min[b_G, b_P] = 56 \text{ mm}$$

- **Overload Factor K_0**

Table of Overload Factors, K_o			
Driven Machine			
Power source	Uniform	Moderate shock	Heavy shock
Uniform	1.00	1.25	1.75
Light shock	1.25	1.50	2.00
Medium shock	1.50	1.75	2.25

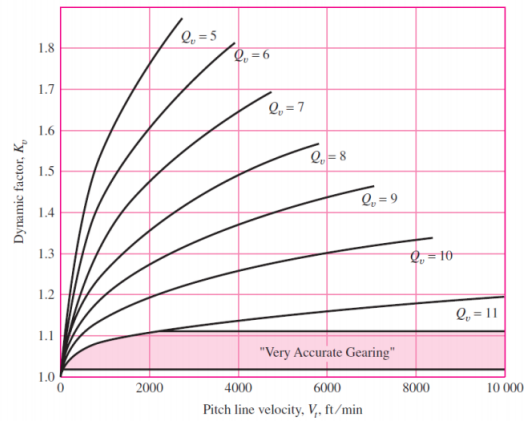
$K_0 = 1$ For a uniform driven and power source machines.

- **Rim-Thickness factor K_B**



$K_B = 1$ as $m_B = 1$

- **Dynamic Factor K_v**



where Q_v is the transmission accuracy-level number and V_t is the peripheral linear speed of the coupling expressed in ft/min, and it is equal to 8.

$$V_t = \omega_3 \cdot r_3 = 2276 \text{ ft/min}$$

where r_3 is the radius of the gear dedendum circle.

$$K_v = 1.17$$

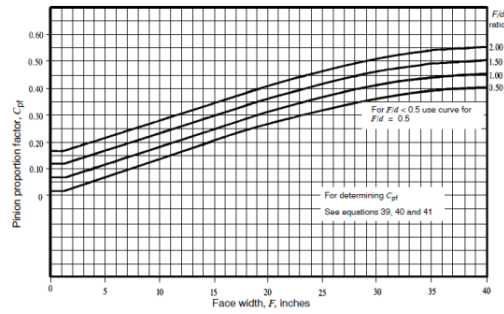
- Load Distribution Factor K_H

$$K_H = 1 + C_{mc}(C_{pf}C_{pm} + C_{ma}C_e), \text{ where } C_e = 1.$$

ATTENTION:

Diagrams are provided here with a tooth width b (F in the plots) expressed in inches

$$C_{mc} = \begin{cases} 1 & \text{for uncrowned teeth} \\ 0,8 & \text{for crowned teeth} \end{cases}$$



where d is the pitch circle diameter, b is the face width.

$$C_{mc} = 1 \text{ for uncrowned teeth}$$

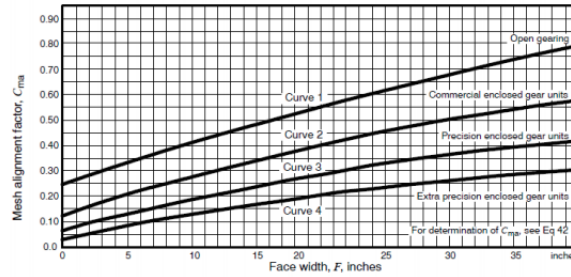
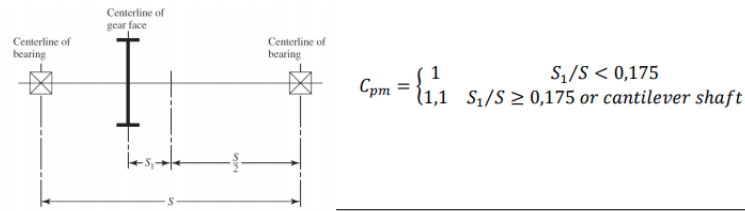
For C_{pf}

$$F = 2,2 \text{ inches } \frac{F}{d} = 0,64 \xrightarrow{\text{Graphically}} C_{pf} = 0,04$$

$$\text{for } C_{mp}: S = a + b + c = 208,75mm$$

$$S_1 = a + b - \frac{S}{2} = 45,125$$

$$\frac{S_1}{S} = 0,216 \xrightarrow{\text{less than } 0,175} C_{mp} = 1,1$$



Graphically $C_{ma} = 0,161$ Finally $K_H = 1,205$

- Size Factor K_S

Number of Teeth	Y	Number of Teeth	Y
12	0.245	28	0.353
13	0.261	30	0.359
14	0.277	34	0.371
15	0.290	38	0.384
16	0.296	43	0.397
17	0.303	50	0.409
18	0.309	60	0.422
19	0.314	75	0.435
20	0.322	100	0.447
21	0.328	150	0.460
22	0.331	300	0.472
24	0.337	400	0.480
26	0.346	Rock	0.485

$$K_S = 0,843(b \cdot m_t \sqrt{Y})^{0,0535}$$

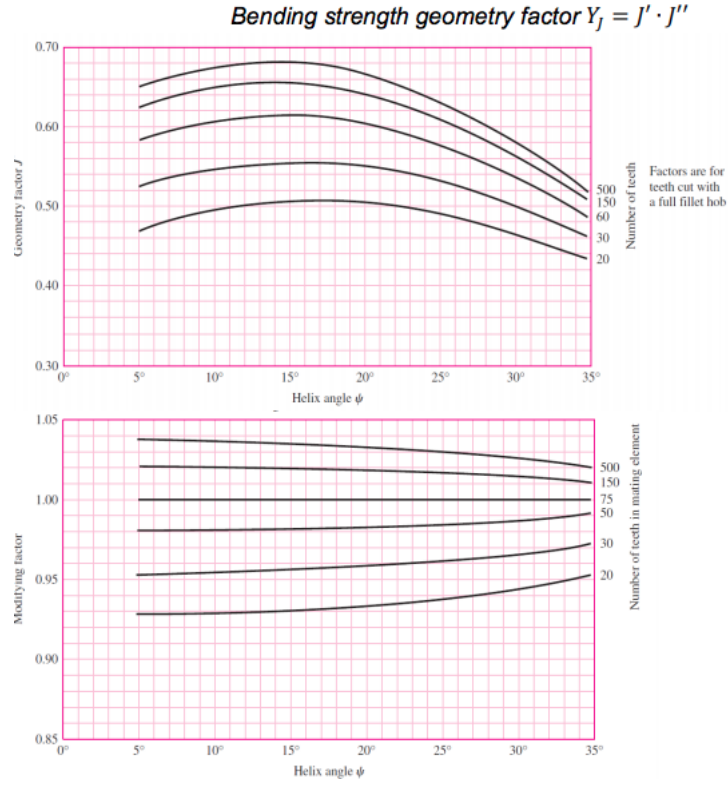
Express b and mt in mm

For helical gears

$$z' = \frac{z}{\cos^3 \psi}$$

Graphically we derive Y and calculate $K_S = 1,095$

- Bending strength geometry factor Y_J



Given the helix angle and the number of teeth of the mached elements:

$$J' = 0,51 \quad J'' = 0,93$$

and thus $Y_J = 0,4743$

As a final step and using equation(17):

$$\sigma_{max,Fatigue} = 221.946 \text{ MPa}$$

4.2 Bending Safety Factor

$$S_F = \frac{\sigma_{FP} \cdot Y_N}{\sigma_{max, Fatigue} \cdot Y_\theta \cdot Y_Z} \quad (18)$$

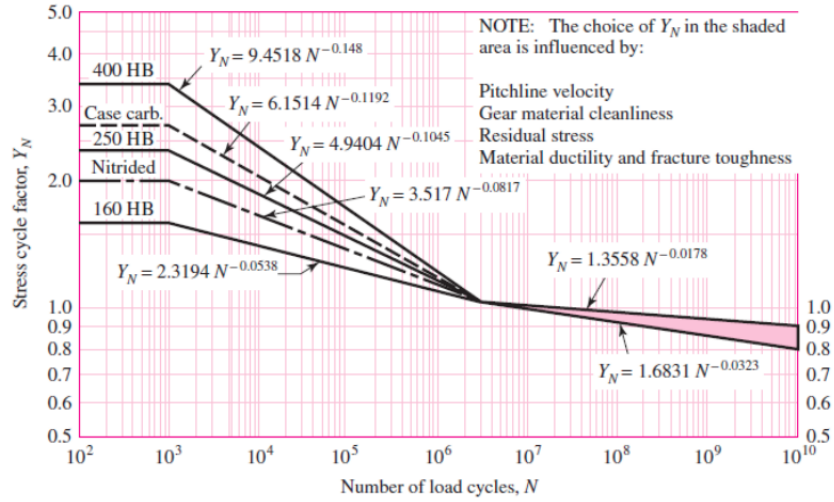
where:

$$\sigma_{FP}(S_{f, lim}) = 860 \text{ MPa}$$

- **Stress cycle life factor Y_N**

For the calculation of Y_N , Graphically or by using the formula:

$$Y_N = 1.3558 \cdot N^{-0.0178}$$



We find out

$$Y_N = 0,97677$$

- **Temperature factor Y_θ**

$$Y_\theta = 1$$

for temperature up to 120 °C

- **Reliability factor Y_Z** // $Y_Z = 1$ for a reliability of 0,99. And thus $S_F = 3,78$

4.3 Maximum gear contact (pitting resistance) stress equation in fatigue

$$\sigma_{max, pitting} = Z_E \sqrt{F_t K_0 K_V K_S \cdot \frac{K_H}{b \cdot d_P} \cdot \frac{Z_R}{Z_I}} \quad (19)$$

- **Elastic coefficient Z_E**
 $Z_E = 191 \sqrt{\text{MPa}}$
- **Surface strength geometry Z_I**

$$\begin{cases} r_{bP} = r_P \cos[\phi_t] \\ r_{bG} = r_G \cos[\phi_t] \end{cases}$$

where r_P e r_G are the pitch radii of the pinion **P** (the smaller gear) and the mating gear **G** respectively while r_{bP} e r_{bG} are the base radii of **P** and **G**. ϕ_t is the transverse pressure angle.

$$\begin{cases} Z_A = \min \left[\sqrt{(r_P + a)^2 - r_{bP}^2}, (r_P + r_G) \sin[\phi_t] \right] \\ Z_B = \min \left[\sqrt{(r_G + a)^2 - r_{bG}^2}, (r_P + r_G) \sin[\phi_t] \right] \\ Z = Z_A + Z_B - (r_P + r_G) \sin[\phi_t] \end{cases} \Rightarrow m_N = \frac{p_n \cos[\phi_n]}{0,95 \cdot Z},$$

Where

- m_N is the load sharing ratio,
- $p_n = \pi \cdot m_n$ is the normal pitch (with m_n the normal module),
- $a = m_n$ is the addendum,
- ϕ_n is the normal pressure angle,
- ϕ_t is the transverse pressure angle with $\tan[\phi_t] = \frac{\tan[\phi_n]}{\cos[\psi]}$.

$$Z_I = \begin{cases} \frac{\cos[\phi_t] \sin[\phi_t]}{2m_N} \cdot \frac{m_G}{m_G + 1} & \text{external meshing} \\ \frac{\cos[\phi_t] \sin[\phi_t]}{2m_N} \cdot \frac{m_G}{m_G - 1} & \text{internal meshing} \end{cases}$$

where $m_G = n_P/n_G = d_G/d_P$ is the gear ratio.

Starting from the equation provided above:

$$r_{bP} = 40.688 \text{ mm}$$

$$P_n = 12.566 \text{ mm}$$

$$r_{bG} = 122.064 \text{ mm}$$

$$\rightarrow Z_A = 24.472 \text{ mm}; Z_B = 56.34 \text{ mm}$$

$$Z = 19.494 \text{ mm}$$

$$\text{Next: } m_N = \frac{p_n \cos(\phi_n)}{0,95 \cdot Z} = 0,6376$$

- $Z_I = 0,194$ external mashing

- $Z_I = 0,3381$ internal mashing

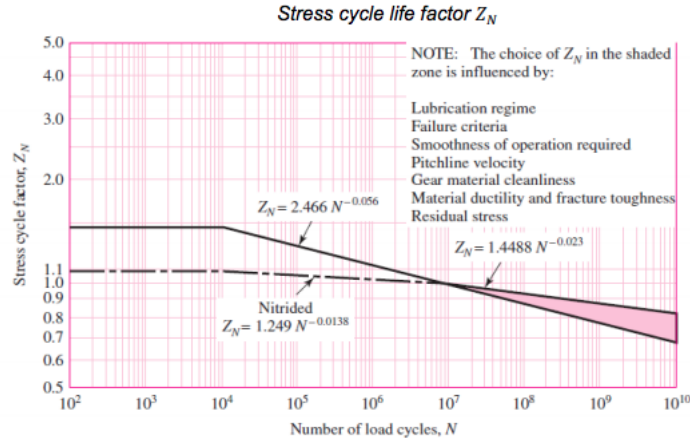
Eventually,

$$\sigma_{max,pitting} = 70.25 \text{ MPa}$$

4.4 Wear safety factor (Hertzian contact)

$$S_H = \frac{\sigma_{HP}}{\sigma_{max,pitting}} \frac{Z_N Z_W}{Y_\theta Y_Z} \quad (20)$$

where $\sigma_{HP}(S_c) = 1470 \text{ MPa}$



Graphically,

Stress cycle life factor $Z_N = 0,948$

$Z_W = 1 + A' \cdot (m_G - 1) = 1$ as A' is zero

Finally,

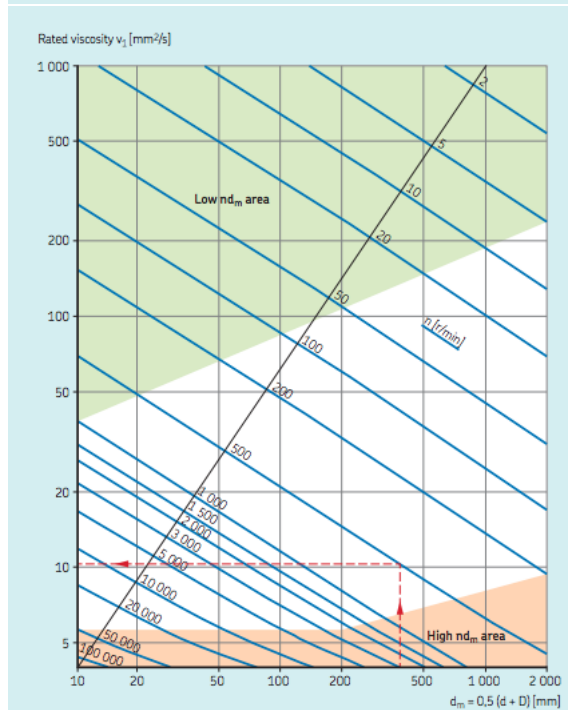
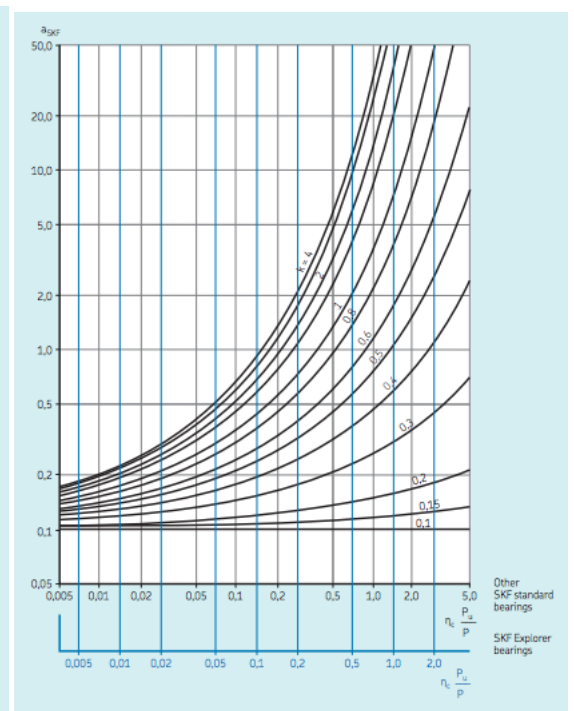
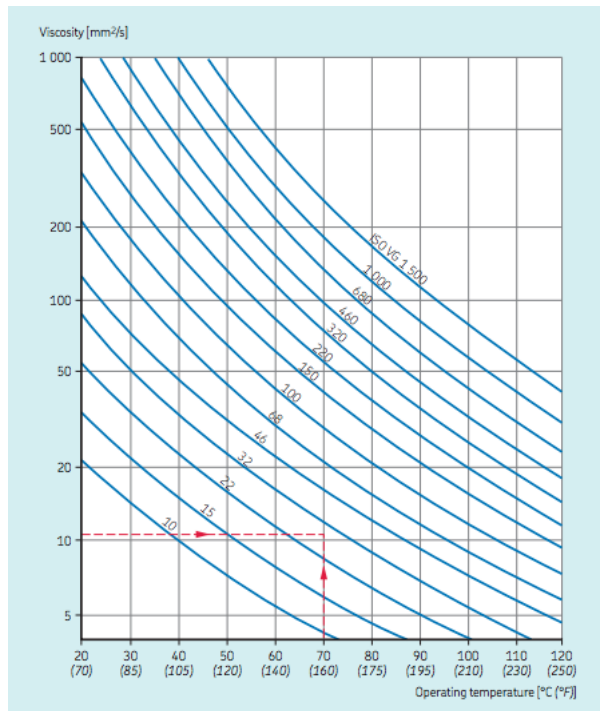
$$S_H = 19,83$$

5 Bearings Life Investigation

5.1 Fatigue Life Bearing

First we need to identify the lubricant used starting from the output shaft where $K = \frac{v}{v_1} = 1$ is given.

We know the value of the viscosity which is 80 we conclude that the Lubricant used is ISO 220 (graphical deduction from the graph of viscosity versus average diameter



For Bearings located in positions B,D,F we use the following method to calculate the dynamic and equivalent loads.

Equivalent dynamic bearing load	$F_d/F_r \leq e \rightarrow P = F_r$	Face-to-face or back-to-back arrangement $F_d/F_r \leq e \rightarrow P = F_r + Y_1 F_a$ $F_d/F_r > e \rightarrow P = 0,67 F_r + Y_2 F_a$ Tandem arrangement ²⁾ $F_d/F_r \leq e \rightarrow P = F_r$ $F_d/F_r > e \rightarrow P = 0,4 F_r + Y F_a$
	$F_d/F_r > e \rightarrow P = 0,4 F_r + Y F_a$ ¹⁾	
For additional information → page 91		
Equivalent static bearing load	$P_0 = 0,5 F_r + Y_0 F_a$ ¹⁾	Face-to-face or back-to-back arrangement $P_0 = F_r + Y_0 F_a$ $P_0 < F_r \rightarrow P_0 = F_r$ Tandem arrangement ²⁾ $P_0 = 0,5 F_r + Y_0 F_a$
	$P_0 < F_r \rightarrow P_0 = F_r$	
For additional information → page 105		

Elaborated in the table, we find out that for these bearing the ratio of axial and radial forces is less than e, and thus the equivalent dynamic load is calculated as $P = F_r + Y_1 F_a$ where Y_1 is extracted from catalogue. While the static load is calculated as shown in the section below.

For Bearings A,C,E the equivalent loads are calculated according to:

	Single row, high-capacity and single row full comple- ment bearings	Double row full complement bearings
Minimum load For additional information → page 106	$F_{rm} = k_f \left(6 + \frac{4n}{n_r} \right) \left(\frac{d_m}{100} \right)^2$	
Equivalent dynamic bear- ing load For additional information → page 91	Non-locating bearings $P = F_r$	
	Locating bearings $F_d/F_r \leq e \rightarrow P = F_r$ $F_d/F_r > e \rightarrow P = 0,92 F_r + Y F_a$ F_a must not exceed $0,5 F_r$	$F_d/F_r \leq 0,15 \rightarrow P = F_r$ $F_d/F_r > 0,15 \rightarrow P = 0,92 F_r + 0,4 F_a$ F_a must not exceed $0,25 F_r$
Equivalent static bearing load For additional information → page 105	$P_0 = F_r$	

$$F_r = \sqrt{R_x^2 + R_y^2}$$

For these bearing $F_r = P_0 = P$ In each Bearing, knowing the calculated the kind of lubricant we can derive the viscosity v and deriving v_1 from the average

diameter of each bearing which is calculated by:

$$d_{avg} = \frac{D + d}{2}$$

After which we will calculate K as their ratio.

Using the diagram for a_{sk} we calculate its value.

$$L_{10m} = a_1 a_{skf} \left(\frac{C}{P}\right)^p \quad (21)$$

$$L_{10mh} = \frac{10^6}{60 \cdot n} L_{10m} \quad (22)$$

Bearings	C_0	C
A:NU 206 ECP	36500	44000
B:30206 DF	88000	85700
C:NU 209 ECP	69500	69500
D:32011 X/DF	232000	170000
E:NU 2210 ECP	90000	90000
F:32011 X/DF	232000	170000

Table 10: C and C_0

Bearings	e	$n_c \frac{P_u}{P}$	P	P_0	P_U	K	a_{skf}	L_{10m}	L_{10h}
A:NU 206 ECP	-	0,846	2924	2924	4500	4,44	50	420416	4671294
B:30206 DF	0,37	0,465	11399,254	11399,254	9650	3,5	10	8324	92492
C:NU 209 ECP	-	0,67	6614,215	6614,215	8150	2,28	10	25410	847033
D:32011 X/DF	0,4	0,9	15749,246	15572	26000	2,66	35	97281,86	3242728,875
E:NU 2210 ECP	-	0,62	10051,9	10051,9	11400	1	2,8	4173,23	417323
F:32011 X/DF	0,4	0,52	27916	27286,616	26000	1	2	824,74	82474,99

Table 11: Summary for all data concerning bearing analysis

5.2 Static Failure

The Static safety factor is calculated as follows:

$$SF = \frac{C_0}{P_0} \quad (23)$$

$$SF_A = 24,96$$

$$SF_B = 7,71$$

$$SF_C = 19,352$$

$$SF_D = 14,98$$

$$SF_E = 17,4$$

$$SF_F = 8,5$$

6 Table For Key Calculation Results

	VALUE	UNIT OF MEASUREMENT
SHAFT ANALYSIS		
$R_{x,C}$	4480,434	N
$R_{y,C}$	-12446,57	N
$R_{x,D}$	-1999,865	N
$R_{y,D}$	-12590,09	N
$R_{z,C}$	0	N
$R_{z,D}$	1765,48	N
$M_x(z=V1)$	745,96	Nm
$M_x(z=V2)$	889,026	Nm
$M_x(z=V3)$	821,70	Nm
$M_x(z=V4)$	745,96	Nm
$M_y(z=V1)$	-65,683	Nm
$M_y(z=V2)$	-128,406	Nm
$M_y(z=V3)$	-98,889	Nm
$M_y(z=V4)$	118,5	Nm
$N(z=V1)$	4237,01	N
$N(z=V2)$	4237,01	N
$N(z=V3)$	4237,01	N
$N(z=V4)$	4237,01	N
$M_{b\text{tot,max}}$	1045,037	Nm
$M_{t\text{tot,max}}$	687,546	Nm
SF (static)	15,42	-
$\sigma_{a,b \text{ nom}} (z=m.c.s.)$	47,74	MPa
$\sigma_{m,b \text{ nom}} (z=m.c.s.)$	0	MPa
$\sigma_{a,N \text{ nom}} (z=m.c.s.)$	0	MPa
$\sigma_{m,N \text{ nom}} (z=m.c.s.)$	1,72	MPa
$\tau_{a,t \text{ nom}} (z=m.c.s.)$	0	MPa
$\tau_{m,t \text{ nom}} (z=m.c.s.)$	19,93	MPa
$K_{f,b}$	2,32	
$\sigma_{a,b \text{ eff}} (z=m.c.s.)$	110,756	MPa
$\sigma_{m,b \text{ eff}} (z=m.c.s.)$	52885	MPa
$K_{f,N}$	2,232	
$\sigma_{a,N \text{ eff}} (z=m.c.s.)$	0	MPa
$\sigma_{m,N \text{ eff}} (z=m.c.s.)$	0	MPa
$K_{f,T}$	1,528	

$\tau_{a,1\text{eff}} (z=m.c.s.)$	0	MPa
$\tau_{m,1\text{eff}} (z=m.c.s.)$	52,74621	MPa
$\sigma_{a,eqP} (z=m.c.s.)$	110,756	MPa
$\sigma_{m,eqP} (z=m.c.s.)$	52,858	MPa
SF _f (fatigue)	3,969	-

• M.c.s. Most critical section

	VALUE	UNIT OF MEASUREMENT
GEAR ANALYSIS - BENDING		
$\frac{F_t}{b \cdot m_t \cdot Y_f}$	143,7955	MPa
$\sigma_{\max, \text{Fatigue}}$	221,946	MPa
S_F	3,7847	-
GEAR ANALYSIS - PITTING		
$\frac{F}{b} \cdot \frac{Z_E \sqrt{t}}{d_P} \cdot \frac{1}{Z_I} \cdot \dots$	56,539	MPa
$\sigma_{\max, \text{pitting}}$	70,25168	MPa
S_H	19,83667	-

	VALUE	UNIT OF MEASUREMENT
BEARING ANALYSIS: LOAD CALCULATION		
$F_{r,A}$	2924,42	kN
$F_{a,A}$	0	kN
$F_{r,B}$	6950,50	kN
$F_{a,B}$	2471,53	kN
P_A	2924,42	kN
P_B	11399,254	kN
$F_{r,C}$	6614,215	kN
$F_{a,C}$	0	kN
$F_{r,D}$	12747,93	kN
$F_{a,D}$	1765,48	kN

P_C	6614,215	kN
P_D	15749,246	kN
F_{r,E}	10091,9	kN
F_{a,E}	0	kN
F_{r,F}	17208,2	kN
F_{a,F}	6299,01	kN
P_E	10091,9	kN
P_F	27916,517	kN

	VALUE	UNIT OF MEASUREMENT
BEARING ANALYSIS: CORRECTED RATING LIFE		
ISO VG of the selected oil	220	mm ² /s
a_{skf,A}	50	-
a_{skf,B}	10	-
a_{skf,C}	10	-
a_{skf,D}	35	-
a_{skf,E}	2,8	-
a_{skf,F}	2	-
L_{10mh,A}	4671294	Hours
L_{10mh,B}	92492,27	Hours
L_{10mh,C}	847032,1	Hours
L_{10mh,D}	3242728,875	Hours
L_{10mh,E}	417323,2	Hours
L_{10mh,F}	82474,99	Hours